ENGINEERING DECISION SUPPORT

SYNTHETIC UNIT

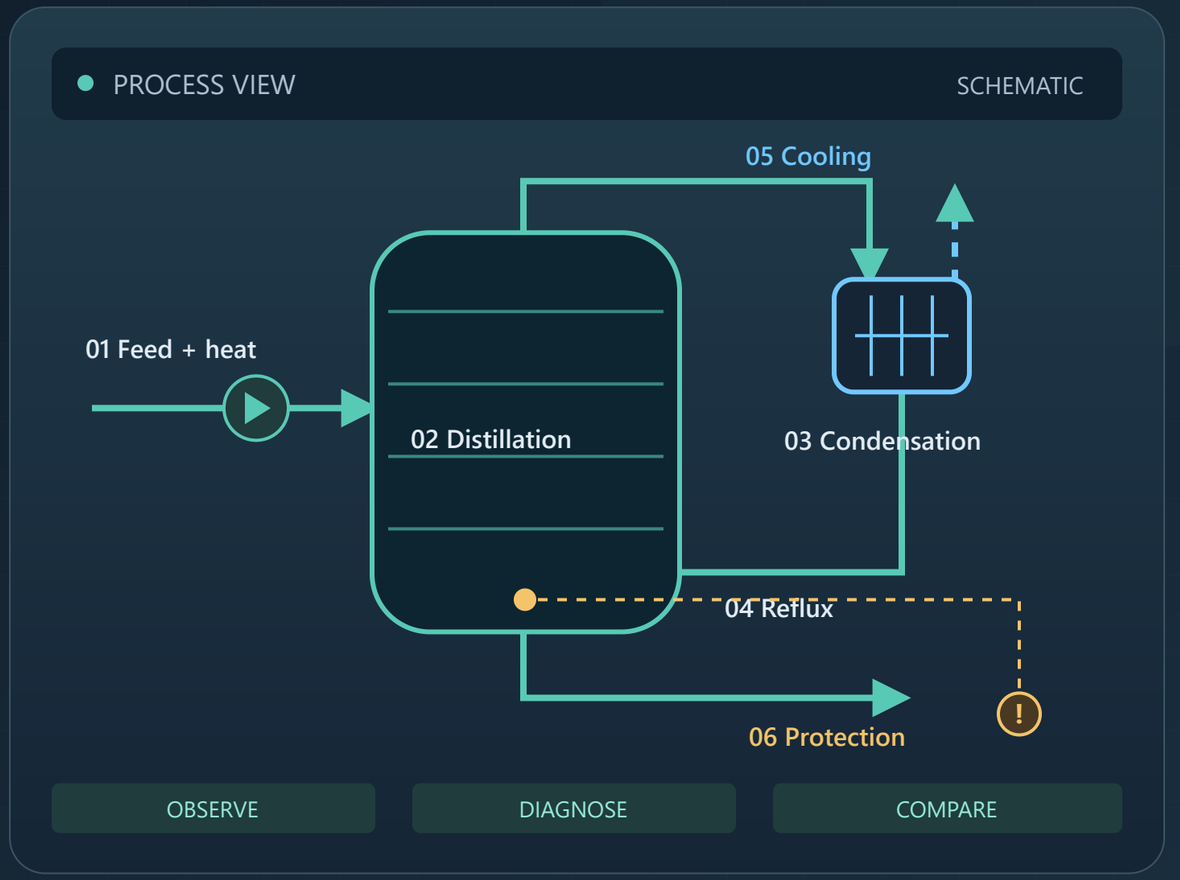
REFINERY

See the signals.

Compare the consequences.

Six loops. Observed evidence.
Explainable hypotheses. What-if forecasts.

CPU DEMO HUMAN REVIEW



Synthetic process simulation · No equipment actuation

AMD Token Factory integration pending

Signals need a decision context.

A temperature rise is an observation. Deciding what to inspect next needs a connected view of the process.

01

See the unit

Follow temperatures, pressures, alarms, and nominal references across six logical loops.

02

Inspect the evidence

Review competing diagnostic hypotheses and the observations behind them.

03

Compare consequences

Explore conditional action forecasts before a human review.

Demonstration scope: a synthetic refinery unit, using simulated observations.

From observations to review.

One calculation connects a process scenario to an inspectable engineering workflow.

SIMULATE

Physical model

Nominal and disturbed trajectories

→

OBSERVE

Available signals

Temperatures, pressures, protection

→

DIAGNOSE

Rank hypotheses

Evidence and similarity scores

→

COMPARE

What-if forecasts

Assumptions and action consequences

Reviewable output

Persisted calculation history, charts, notes, and exports.

Clear information boundary

Hidden simulator truth is separate from the diagnostic observation view.

Six loops. One shared view.

- 01 Feed + heating
- 02 Distillation
- 03 Condensation
- 04 Reflux return
- 05 Cooling
- 06 Independent protection

Built for inspection

Expand all equipment signals. Align charts in time. Review alarms and engineering notes.

Built for presentation

Dedicated video wall, loop summaries, histograms, and shared-time trends.

EN / RU · Light / dark · Configurable labels

> Loop summaries describe available observations; they do not invent unmeasured utility data.

Investigate a cooling disturbance.

The user selects a synthetic scenario. Diagnosis then works from the resulting observations.

1

Run + inspect

Start a cooling-loss scenario. Follow pressure and temperature against the nominal reference.

2

Review hypotheses

Open ranked hypotheses, compare their evidence, and inspect the selected analysis time.

3

Compare actions

Compare hold, restore cooling, reduce heating, and reduce feed under stated assumptions.

Conditional model forecasts support review. The prototype issues no equipment commands.

Make the limits visible.

What the prototype demonstrates

- A working synthetic process simulation and operator interface.
- Observation-based hypothesis ranking and action forecasts.
- Persisted results, engineering review, and report exports.
- A limited A/B/C synthetic model experiment with integrity checks.

What still needs validation

- Similarity scores are not calibrated probabilities.
- Forecasts depend on model assumptions.
- Imported data has unverified provenance.
- Real-plant calibration, authenticated integration, and industrial validation remain future work.

> Missing observations remain missing. Display alarm settings do not change physical protection.

A reproducible CPU demo today.

Implemented stack

Python

NumPy · SciPy · CoolProp

JavaScript · HTML · CSS

Local server and browser workspace.
No API key is required for the CPU demo.

PENDING

AMD Token Factory integration

Account registration and an official endpoint/model are still required.

No AMD inference or GPU execution is claimed.

Proposed track: Intelligent Industry

Run. Inspect. Compare.

REFINERY turns a synthetic disturbance into an evidence-led engineering review.

```
python -m amd.server
```

Open http://127.0.0.1:8765/industry

1 / Run

Create a new calculation with action comparison.

2 / Inspect

Follow signals, evidence, and ranked hypotheses.

3 / Compare

Review conditional forecasts and the video wall.

Public repository, hosted demo, and video publication links will be verified before submission. AMD Token Factory integration remains pending.